Field	Value	Planning note
Project ID	R26-IT-005	
Project name	Automated Water Quality and Contamination Detection	Four linked research components
Plan basis	Proposed work from supplied research descriptions	No completed implementation or evaluation is asserted.
Schedule	To be agreed by the team	Enter real dates in the amber cells on Tasks.
Task status	Planned	Update after confirming actual progress.
Owners	Component assignments from thesis context	Confirm the classification member name on Users.
Dependencies	Suggested sequence within each component	Parallel development is possible after shared schema agreement.
Evidence limits	Visual and microscopy predictions retain uncertainty	Predictions alone do not establish potability, material identity or organism viability.
Source	User-provided research scope and uploaded planner	Unrelated project tasks, users and completion dates have been replaced.

Component / phase	Owner / role	Total tasks	Planned	In Progress	Blocked	Completed
Project Setup	Team	5	5	0	0	0
Contamination Detection	IT22295460	7	7	0	0	0
Microorganism Detection	IT22212122	7	7	0	0	0
Water Quality Classification	IT22211750	6	6	0	0	0
Report and Purification Advisor	IT22193940	8	8	0	0	0
Integration and Deployment	Team	4	4	0	0	0
Testing and Thesis	Team	4	4	0	0	0

Task ID	Task name	Component / phase	Goal	Status	Priority	Owner / role	Planned start	Planned due	Dependency	Expected deliverable	Progress notes
SET-01	Agree research scope and component boundaries	Project Setup	G1	Planned	High	Team				Approved scope and research questions	
SET-02	Review related water-quality research	Project Setup	G1	Planned	Medium	Team			SET-01	Literature comparison and bounded research gaps	
SET-03	Define sample IDs, units and output schema	Project Setup	G1	Planned	Medium	Team			SET-02	Versioned JSON schema with provenance and availability	
SET-04	Create repository and repeatable environments	Project Setup	G1	Planned	Medium	Team			SET-03	Repository, dependency versions and configuration guide	
SET-05	Agree data access, licensing and review process	Project Setup	G1	Planned	Medium	Team			SET-04	Dataset permissions and expert-review protocol	
VIS-01	Define discrete and diffuse visual classes	Contamination Detection	G2	Planned	High	IT22295460			SET-03	Annotation guide separating debris from cloudiness regions	
VIS-02	Collect and review water-sample photographs	Contamination Detection	G2	Planned	Medium	IT22295460			VIS-01	Licensed dataset with capture conditions and sample IDs	
VIS-03	Annotate and split photographic dataset	Contamination Detection	G2	Planned	Medium	IT22295460			VIS-02	Reviewed boxes; sample-level train/validation/test split	
VIS-04	Train YOLO and Faster R-CNN baselines	Contamination Detection	G2	Planned	Medium	IT22295460			VIS-03	Versioned checkpoints under common evaluation conditions	
VIS-05	Develop visual-severity estimation	Contamination Detection	G2	Planned	Medium	IT22295460			VIS-04	Severity rubric and agreement with reviewed labels	
VIS-06	Evaluate detection robustness	Contamination Detection	G2	Planned	Medium	IT22295460			VIS-05	Per-class precision, recall, mAP and latency across capture conditions	
VIS-07	Expose photographic detection API	Contamination Detection	G2	Planned	Medium	IT22295460			VIS-06	Boxes, visual labels, confidence, severity and limitations	
MIC-01	Define supported microscopy labels	Microorganism Detection	G3	Planned	High	IT22212122			SET-03	Taxonomic resolution justified by imagery and reference labels	
MIC-02	Prepare labelled microscopy dataset	Microorganism Detection	G3	Planned	Medium	IT22212122			MIC-01	Imaging metadata, reference labels and specimen-level splits	
MIC-03	Review organism annotations	Microorganism Detection	G3	Planned	Medium	IT22212122			MIC-02	Reviewed boxes and ambiguous/unidentified labels	
MIC-04	Train YOLO and Faster R-CNN baselines	Microorganism Detection	G3	Planned	Medium	IT22212122			MIC-03	Versioned models under common data conditions	
MIC-05	Implement quality and uncertainty handling	Microorganism Detection	G3	Planned	Medium	IT22212122			MIC-04	Insufficient-evidence outcome and confidence fields	
MIC-06	Evaluate microorganism detection	Microorganism Detection	G3	Planned	Medium	IT22212122			MIC-05	Per-class precision, recall, mAP and specimen-held-out analysis	
MIC-07	Expose microscope analysis API	Microorganism Detection	G3	Planned	Medium	IT22212122			MIC-06	Boxes and supported labels; counts limited to analysed fields	
CLS-01	Define parameters and classification target	Water Quality Classification	G4	Planned	High	IT22211750			SET-03	pH, hardness, turbidity, TDS and fluoride units; label provenance	
CLS-02	Prepare tabular water-quality dataset	Water Quality Classification	G4	Planned	Medium	IT22211750			CLS-01	Missingness audit, leakage controls and held-out split	
CLS-03	Train Random Forest and Gradient Boosting	Water Quality Classification	G4	Planned	Medium	IT22211750			CLS-02	Reproducible preprocessing and comparable model runs	
CLS-04	Implement parameter-level flags	Water Quality Classification	G4	Planned	Medium	IT22211750			CLS-03	Versioned thresholds and missing-input handling	
CLS-05	Evaluate classification and calibration	Water Quality Classification	G4	Planned	Medium	IT22211750			CLS-04	Confusion matrix, per-class recall, F1 and calibration results	
CLS-06	Expose water-quality classification API	Water Quality Classification	G4	Planned	Medium	IT22211750			CLS-05	Quality class, parameter flags, confidence and provenance	
RAG-01	Curate approved water-treatment references	Report and Purification Advisor	G5	Planned	High	IT22193940			SET-03	Versioned corpus with source, date and applicability metadata	
RAG-02	Build ingestion, chunking and retrieval	Report and Purification Advisor	G5	Planned	Medium	IT22193940			RAG-01	Retrieval index and relevance-labelled evaluation cases	
RAG-03	Validate and reconcile upstream evidence	Report and Purification Advisor	G5	Planned	Medium	IT22193940			RAG-02	Sample-ID, units, missingness and contradiction checks	
RAG-04	Encode reviewed recommendation rules	Report and Purification Advisor	G5	Planned	Medium	IT22193940			RAG-03	Eligibility, prerequisites and abstention rules	
RAG-05	Implement deterministic calculations	Report and Purification Advisor	G5	Planned	Medium	IT22193940			RAG-04	Documented units and independently checked expected values	
RAG-06	Generate evidence-grounded reports	Report and Purification Advisor	G5	Planned	Medium	IT22193940			RAG-05	Citations, limitations and immutable calculation/status fields	
RAG-07	Evaluate reporting baselines	Report and Purification Advisor	G5	Planned	Medium	IT22193940			RAG-06	Template, ungrounded LLM, basic RAG and hybrid comparison	
RAG-08	Review advice and user understanding	Report and Purification Advisor	G5	Planned	Medium	IT22193940			RAG-07	Expert appropriateness rubric and approved participant protocol	
INT-01	Integrate four component APIs	Integration and Deployment	G6	Planned	High	Team			SET-03	Versioned requests, shared sample identity and provenance	
INT-02	Build report interface and export	Integration and Deployment	G6	Planned	Medium	Team			INT-01	Readable reports with evidence status and cited sources	
INT-03	Implement audit records and failure handling	Integration and Deployment	G6	Planned	Medium	Team			INT-02	Version-aware caching, protected logs and template fallback	
INT-04	Prepare restricted research deployment	Integration and Deployment	G6	Planned	Medium	Team			INT-03	Configuration, rollback procedure and supervised-pilot review gate	
TST-01	Run unit and contract tests	Testing and Thesis	G7	Planned	High	Team			SET-03	Schema, coordinate, unit, rule-boundary and arithmetic checks	
TST-02	Run integration and adversarial tests	Testing and Thesis	G7	Planned	Medium	Team			TST-01	Missing modules, conflicts, timeouts and prompt-injection cases	
TST-03	Execute held-out evaluation protocol	Testing and Thesis	G7	Planned	Medium	Team			TST-02	Separate software verification from research performance	
TST-04	Prepare theses and final demonstration	Testing and Thesis	G7	Planned	Medium	Team			TST-03	Methods, limitations, results when available and reproducibility package	

Goal ID	Goal name	Status	Priority	Start date	Finish date	Connected tasks	Acceptance condition
G1	Project Setup	Planned	High			SET-01, SET-02, SET-03, SET-04, SET-05	Completion requires review of the listed deliverables.
G2	Contamination Detection	Planned	High			VIS-01, VIS-02, VIS-03, VIS-04, VIS-05, VIS-06, VIS-07	Completion requires review of the listed deliverables.
G3	Microorganism Detection	Planned	High			MIC-01, MIC-02, MIC-03, MIC-04, MIC-05, MIC-06, MIC-07	Completion requires review of the listed deliverables.
G4	Water Quality Classification	Planned	High			CLS-01, CLS-02, CLS-03, CLS-04, CLS-05, CLS-06	Completion requires review of the listed deliverables.
G5	Report and Purification Advisor	Planned	High			RAG-01, RAG-02, RAG-03, RAG-04, RAG-05, RAG-06, RAG-07, RAG-08	Completion requires review of the listed deliverables.
G6	Integration and Deployment	Planned	High			INT-01, INT-02, INT-03, INT-04	Completion requires review of the listed deliverables.
G7	Testing and Thesis	Planned	High			TST-01, TST-02, TST-03, TST-04	Completion requires review of the listed deliverables.

Bucket ID	Bucket name
SET	Project Setup
VIS	Contamination Detection
MIC	Microorganism Detection
CLS	Water Quality Classification
RAG	Report and Purification Advisor
INT	Integration and Deployment
TST	Testing and Thesis

Owner / role	Name	Component
IT22295460	W.A.S.P.D. Wijesingha	Contamination Detection
IT22212122	D. Dodanwela	Microorganism Detection
IT22211750	Name to confirm	Water Quality Classification
IT22193940	R.M.N.A. Rathnayake	Report and Purification Advisor
Team	All project members	Shared setup, integration and testing